Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Soller, Babs R. Ph.D.		
Project Title:	Noninvasive Measurement of Blood and Tissue Chemistry		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Previous affiliation was University of Massachusetts Medical School, Worcester, MA, until October 2010 (Ed., 11-9-2010)		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Heard, Stephen (University of Massachusetts Medical School)		
Grant/Contract No.:	NCC 9-58-SMS00403		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Trauma and acute medical problems along with loss of skeletal muscle mass, strength and endurance, are some of the most serious risks facing astronauts during long duration spaceflight. The measurement of two metabolic parameters, muscle pH and oxygen can be applied in both areas. This project examines the hypothesis that near infrared spectroscopy (NIRS) is a platform technology that can provide noninvasive physiologic monitoring in support of multiple NASA needs. The immediate goal is to produce, validate and deliver to NASA-JSC a small, lightweight medical monitor which utilizes NIRS to measure important metabolic parameters. Requisite precision and accuracy will be demonstrated for both male and female subjects of any ethnic origin. The device will perform equally well on the forearm and the thigh, despite gender and weight related differences in fat thickness. The specific aims of this project are: (1) Develop calibration procedures for modified NIRS sensor to measure deep muscle metabolic parameters (tissue pH and oxygen); (2) Validate sensor in exercise protocol and deliver systems to NASA-JSC; (3) Determine values for tissue pH and oxygen that can be used by a smart medical system to indicate shock and assist in guiding treatment; and (4) Evaluate hardware for flight requirements and develop plan to produce flight-ready instrumentation. During year 2 of this project we improved the robustness of our calibration equations by developing methods that simultaneously corrected for spectral interferences from skin pigmentation and fat. We also developed new methodology that allowed us to correct for subject-to-subject differences in muscle optical properties. These innovations considerably increased our ability to accurately measure muscle pH, oxygen and hematocrit across all subjects. We completed development of hardware and software to measure muscle metabolic parameters on the arm (thinner fat layer) and the leg (thicker fat layer). An arm system was delivered to the exercise physiology group at Johnson Space Center (JSC) and was used in a pilot study to develop a pre-EVA handgrip fitness test. The leg system development was completed and is currently being validated against invasive measurements at the University of Massachusetts Medical School, in a knee extension protocol. The system will be delivered to JSC in late spring. The other intended application of the system will be in a Smart Medical System for the assessment and treatment of trauma or injury which results in reduced blood pressure, blood flow or cardiac output. We have begun a study in the UMass Medical School Emergency Department to apply the sensor to subjects at risk for developing shock. We have been able to collect data from 3 subjects who were in shock. These initial results indicate that we were able to detect changes in peripheral perfusion that accompany shock, and that the noninvasive sensor was also able to detect improvements in muscle oxygenation that paralleled standard invasive clinical measurements of improvement (blood lactate). Recruitment into this protocol is much lower than anticipated and we are beginning to explore the possibility of adding another site to the study to improve subject accrual. During frequent trips to Johnson Space Center we have begun conversations with biomedical engineers and medical operations personnel to begin to understand the requirements to proceed to a flight study. Early conversations have indicated a likely path, though it is not clear that funds will be available to “flybadize” the hardware in the near future. We are hoping to test the device as part of the Exercise Countermeasure Program’s planned bed rest study, to validate its application in assessing loss of muscle strength in a space analog system. The NIRS noninvasive metabolic monitor is expected to have many applications for NASA. The system will have additional use on earth for military and civilian personnel treating mass casualties. It can also be used in the hospital, ambulances and helicopters. As part of a Smart Medical System, advanced medical assessment and monitoring may become available to physicians in remote and rural areas, who may not have access to specialist expertise.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will have direct earth based application. The prototype monitors we are developing will have application in emergency response vehicles, emergency rooms, and hospitals. Pre-hospital applications include assessing the severity of shock and triaging multiple casualties, as well as providing a sensor for a smart medical system to guide resuscitation from hemorrhage. In the ICU we expect that this monitor will find application in distinguishing between hemorrhagic and septic shock and helping to assess the effectiveness of sepsis treatments. The direct muscle application of interest to NASA for assessing fitness in space, may be useful to assess success of physical therapy in rehabilitating patients with muscle injury or atrophy. There is also general medical application for the diagnosis of anemia and if small and inexpensive enough, screening world-wide for malnutrition. There is also a possible application for the diagnosis of diabetic foot ulcers. A smaller version of this monitor could find use in the training of elite and weekend athletes.
Task Progress:	There are 3 separate aspects of this program that are being pursued in parallel: 1) enhancement of calibration equation accuracy, 2) development of hardware for leg measurement and 3) application of the system to assess trauma patients and guide treatment. 1) Calibration Equations We developed a novel method to correct transdermal spectra for interference from overlying layers of skin and fat. This method employs a specially designed sensor and novel mathematics. We demonstrated this method on phantoms and a paper was published in Optics Letters. We also demonstrated this for the measurement of hematocrit on human subjects. We found that we also needed to correct for variation in muscle optical properties between subjects. We developed a new mathematical method for this and demonstrated it in a phantom study, and on pH measurement of human subjects. A paper has been written for Applied Spectroscopy on this technique. An invention disclosure will be prepared shortly. We also investigated methods to calculate the absolute concentration of capillary hemoglobin and deoxygenated hemoglobin from NIRS. This novel method also allows us to track changes in tissue water concentration and blood volume over time. 2) Leg System We had to significantly redesign the optical system to get sufficient light from leg muscle because the thick fat layer on the leg tends to reflect most of the light away. The leg system increases the lamp power from 8W to 50W. This required a total redesign of the power management system. The leg system also required a complete redesign of the fiber optic cable. The source detector spacing was increased from 30 mm to 40 mm and the diameter of the fiber bundle in lamp cable also increased. Initial testing during knee extension dynamometry was completed using the Paratrend invasive sensors as references, however many of the sensor broke during the study, and many of the spectra were collected without proper reference measurements, making them unsuitable for calibration equation development. The leg

	calibration/validation study must be repeated using blood samples as a reference. 3) Trauma Application The Emergency Department study began earlier than planned. We did complete a control study and collected data on 3 patients in shock. Subject recruitment has been slow and difficult. We recently obtained IRB permission to do the study without consent if the family is not available. This should help recruitment, but we will most likely have to involve an additional site to collect sufficient data.
Bibliography Type:	Description: (Last Updated: 03/22/2019)
Abstracts for Journals and Proceedings	Soller BR, Soyemi O, Landry M, Shear M, Wu J, Hagan, RD. "Validation of a new NIRS method for measuring muscle oxygenation during rhythmic handgrip exercise." American College of Sports Medicine 53rd annual meeting, Denver, CO, May 31-June 3, 2006. Med Sci Sports Exerc. 2006 May;38(5) Suppl:S248-S249. , May-2006
Abstracts for Journals and Proceedings	Wu J, Soyemi O, Walz M, Yang Y, Soller B. "Preliminary use of noninvasive, NIRS-measured tissue perfusion in critically ill patients." Crit Care Med. 2005 December. Crit Care Med. 2005 December;33 (12-Suppl):A126. , Dec-2005
Articles in Peer-reviewed Journals	Yang Y, Landry MR, Soyemi OO, Shear MA, Anunciacion DS, Soller BR. "Simultaneous correction of the influence of skin color and fat on tissue spectroscopy by use of a two-distance fiber-optic probe and orthogonalization technique." Opt Lett. 2005 Sep 1;30(17):2269-71. PMID: 16190440 , Sep-2005
Articles in Peer-reviewed Journals	Yang Y, Soyemi OO, Landry MR, Soller BR. "Influence of a fat layer on the near infrared spectra of human muscle: quantititate analysis based on two-layered Monte Carlo simulations and phantom experiments." Opt Express. 2005 Mar 7;13(5):1570-9. PMID: 16044624 , Mar-2005
Awards	Soller B. "Babs Soller named to the US Army Integrated Product Team developing Advanced Capabilities for the Combat Medic." Jan-2006
Awards	"Babs Soller promoted to Full Professor." Jan-2006
Awards	"Olusola Soyemi promoted from Instructor to Assistant Professor." Jan-2006
Awards	"Stephen Heard named President of the American Society of Critical Care Anesthesiologists." Jan-2006
Papers from Meeting Proceedings	Yang Y, Shea, MA, Soyemi OO, Soller BR. "Effect of skin and fat layers on the spatial sensitivity profile of continuous wave diffuse reflectance near-infrared spectra." Smart Medical and Biomedical Sensor Technology III symposium, Boston, MA, Oct 24, 2005. Proc SPIE. 2005 Nov;6007:60070, M1-6007M9. http://dx.doi.org/10.1117/12.630683 , Nov-2005
Papers from Meeting Proceedings	Yang Y, Soyemi OO, Landry MR, Soller BR. "Influence of a fat layer on the near infrared spectra of human muscle: quantititate analysis based on two-layered Monte Carlo simulations and phantom experiments." Optical Diagnostics and Sensing V, San Jose, CA, January 2005. Proc SPIE. 2005 Mar;5702:104-12. http://dx.doi.org/10.1117/12.585256 , Mar-2005
Papers from Meeting Proceedings	Soyemi O, Shear M, Landry M, Anunciacion D, Soller B. "In-vivo, noninvasive measurement of muscle pH during exercise using near infrared spectroscopy." Smart Medical and Biomedical Sensor Technology III symposium, Boston, MA, Oct 24, 2005. Proc SPIE. 2005 Nov;6007:60070, N1-60070N8. http://dx.doi.org/10.1117/12.630646 , Nov-2005
Papers from Meeting Proceedings	Shear MA, Soyemi OO, Landry M, Soller BR. "Multivariate calibration with slowly responding reference measurements." Smart Medical and Biomedical Sensor Technology III Symposium, Boston, MA, Oct 24, 2005. Proc SPIE. 2005 Nov;6007:60070, O1-60070O8. http://dx/doi.org/10.1117/12.630701 , Nov-2005
Patents	11/113,347. Patent 2006. Apr-2006 Phillipps P, Soller BR, Parker M. "Spectroscopic System for Reflectance Measurements."
Patents	60/674,379. Patent April 2006. Apr-2006 Yang Y, Soller BR, Soyemi O, Shear M. "Systems and Methods for Correcting Optical Reflectance Measurements."